Revisiting the Estimation of Unigram Surprisal in Computational Psycholinguistics

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Abstract

Contextual and unigram surprisal are standard predictors of reading time in computational psycholinguistics. While contextual surprisal is typically derived from neural language models, unigram surprisal is usually obtained from independent corpus counts. Although both quantities are deterministic functionals of a single underlying distribution—capturing conditional and marginal structure respectively—this practice breaks their probabilistic link and may introduce estimation error from the mismatch. We propose three principles for estimating unigram surprisal in psycholinguistic research, including the requirement that both quantities be derived from the same language model. We operationalize this principle by introducing a Monte Carlo estimator of unigram surprisal from a language model, along with a marginalized variant that has lower variance empirically. When unigram surprisal is re-estimated under this unified framework, we find evidence that predictors change in their predictive contribution to reading time. We also confirm prior findings that frequency can be a stronger predictor of processing effort than contextual surprisal under this unified framework.

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1 Introduction

Human sentence processing is incremental (Marslen-Wilson, 1975; Tanenhaus et al., 1995; Rayner and Clifton Jr, 2009) and predictive (Altmann and Kamide, 1999; DeLong et al., 2005): comprehenders do not wait until a sentence is complete before interpreting it, but instead build partial representations and continuously generate expectations about upcoming input. Contextual surprisal—the negative log probability of a linguistic unit given its preceding context as derived from language models (LMs)—quantifies the degree to which incoming content aligns with these predictions. Surprisal has been found to robustly predict observed processing effort in humans across paradigms and languages (Hale, 2001; Wilcox et al., 2020; Kuribayashi et al., 2022; Shain et al., 2024), lending broad empirical support to expectation-based theories of comprehension (Levy, 2008; Kuperberg and Jaeger, 2016; Ryskin and Nieuwland, 2023). Separately, the frequency of a unit in a language also shapes how quickly it is processed: more frequent units are recognized faster, independent of their contextual fit (Hare et al., 1995; Bybee, 1998; Shain, 2024). This effect is captured by unigram surprisal—the negative log probability of a unit after marginalizing over all contexts. While estimating contextual surprisal has benefited greatly from advances in neural LMs (Goodkind and Bicknell, 2018; Wilcox et al., 2020; Oh and Schuler, 2023a; de Varda and Marelli, 2023; Wilcox et al., 2023a), unigram surprisal has received comparatively little methodological attention. Typically, it is derived from small-data corpora or lexical databases (Speer, 2022).

The	soup	was	seasoned	with	**salt**	**coriander**
21.75	17.99	6.77	13.66	2.89	2.96	8.58
4.22	16.11	7.24	18.24	7.14	14.62	20.10
16.04	*21.41*	*13.54*	*28.38*	*13.14*	*19.63*	*31.52*
−17.53	**−1.88**	**0.47**	**4.57**	**4.25**	**11.66**	**11.52**
−5.72	***3.42***	***6.77***	***14.71***	***10.25***	***16.67***	***22.94***

Figure 1: Contextual surprisal, unigram surprisal (WF / MMC), and PMI (WF / MMC) for each word in the prefix *The soup was seasoned with* and two target words ***salt*** and ***coriander***. Unigram surprisal is obtained from two sources: the `wordfreq` package (WF; Speer, 2022) and a marginalized Monte Carlo estimate (MMC; italicized) from mGPT (Shliazhko et al., 2024), which is also used for contextual surprisal.

In a sense made specific by Opedal et al. (2024), contextual surprisal and unigram surprisal are correlated in naturalistic text. That being the case, modern studies on surprisal have controlled for un-

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igram surprisal (Fernandez Monsalve et al., 2012; Smith and Levy, 2013; Shain, 2019; Wilcox et al., 2023b; Shain, 2024). However, controlling for unigram surprisal is only meaningful when both estimates come from the same distribution. In the example in Fig. 1, we calculate contextual surprisal from the language model mGPT (Shliazhko et al., 2024), while we calculate unigram surprisal with two methods: estimates from the `wordfreq` package (Speer, 2022) and estimates obtained with a marginalized Monte Carlo (MMC) estimator over the same language model as contextual surprisal. We observe that the pointwise mutual information (PMI) (the difference between unigram and contextual surprisal) is nearly identical for *salt* and *coriander* under `wordfreq`, but diverges substantially under the MMC estimator. This illustrates the estimation error that can arise from a mismatch in unigram and contextual surprisal distributions.

Motivated by this discrepancy, we put forward a set of three principles to guide approaches to estimating unigram surprisal for research on sentence processing and psycholinguistics more broadly: the best estimate principle, the cognitive plausibility principle, and the consistent distribution principle (see Fig. 2; §5). Though we argue that all three principles should guide unigram surprisal estimations, we note that no existing approach to estimating unigram surprisal satisfies the consistent distribution principle. Such an approach would eliminate the potential for error introduced by mismatched estimation procedures and allow practitioners to leverage the data scale of large LMs for unigram surprisal estimation. To that end, we introduce a Monte Carlo estimator that estimates unigram frequency from the same distribution as an LM by sampling strings from the model and calculating a unit's empirical frequency across these samples. We go on to derive a marginalized variant of the estimator, which we empirically demonstrate to be a more efficient estimator; that is, given the same number of samples, the marginalized variant will obtain lower variance estimates than the naive Monte Carlo estimator.

In the empirical portion of the article, we study several predictors of reading times across eleven languages, as in Opedal et al. (2024), and directly compare models of reading time fit using unigram surprisal values derived from the `wordfreq` lexical database (Speer, 2022) and values from our MMC estimator. Like Opedal et al. (2024), we find that unigram surprisal is a stronger predictor of reading time than contextual surprisal; however, we find that the predictive power of some predictors significantly changes when unigram surprisal is calculated with our MMC estimator. Though our empirical findings focus specifically on sentence processing, they indicate that the estimation of unigram surprisal should be reconsidered more broadly in computational psycholinguistics.

Principles

Best estimate principle. Unigram surprisal estimates should closely approximate the marginal probability of units according to the true distribution over units that generated the corpus used for estimation.

Cognitive plausibility principle. Estimates of unigram surprisal should aim to approximate the marginal probability of units according to the human LM.

Consistent distribution principle. Unigram and contextual surprisal estimates should be derived from the same underlying LM.

Figure 2: Principles for estimating unigram surprisal.

2 Language Models

Preliminaries. An alphabet is a finite, non-empty set. Let Σ be an alphabet of tokens. We denote elements $\sigma \in \Sigma$ by lowercase Greek letters, and strings $\boldsymbol{\sigma} = \sigma_1 \cdots \sigma_T$ by bold lowercase Greek letters, where σ_t denotes the tth token of $\boldsymbol{\sigma}$. Furthermore, let $\mathcal{W}$ be an inventory of words, i.e., a non-empty set which may be finite or countably infinite (Snæbjarnarson et al., 2026; Kiegeland et al., 2026). We denote elements of $\mathcal{W}$ by w. Σ^* denotes the set of all finite strings formed by concatenating tokens in Σ, including the empty string ε. We write $|\boldsymbol{\sigma}| = T$ for the length of $\boldsymbol{\sigma} = \sigma_1 \cdots \sigma_T$. A **prefix** of $\boldsymbol{\sigma}$ is an initial substring; specifically, $\boldsymbol{\sigma}_{<t}$ consists of its first $t-1$ tokens. We write $\boldsymbol{\sigma}' \preceq \boldsymbol{\sigma}$ to indicate that $\boldsymbol{\sigma}'$ is a prefix of $\boldsymbol{\sigma}$, and $\boldsymbol{\sigma}' \prec \boldsymbol{\sigma}$ for a *strict* prefix. Concatenation of $\boldsymbol{\sigma}$ and $\boldsymbol{\sigma}'$ is written $\boldsymbol{\sigma}\boldsymbol{\sigma}'$. Finally, we introduce a distinguished end-of-string symbol $\text{EOS} \notin \Sigma$ and define the augmented alphabet $\overline{\Sigma} \stackrel{\text{def}}{=} \Sigma \cup \{\text{EOS}\}$.

Language models. Given an alphabet Σ, a **language model** (**LM**) is a probability distribution over Σ^*. We denote a language model as p_Σ and use X to denote a random variable taking values

in Σ^* distributed according to p_Σ. Given a string $\boldsymbol{\sigma} \in \Sigma^*$, its prefix probability is:[1]

$$p_{\vec{\Sigma}}(\boldsymbol{\sigma}) \stackrel{\text{def}}{=} \mathop{\mathbb{P}}_{X \sim p_\Sigma}[\boldsymbol{\sigma} \preceq X] = \sum_{\boldsymbol{\sigma}' \in \Sigma^*} \mathbb{1}\{\boldsymbol{\sigma} \preceq \boldsymbol{\sigma}'\}\, p_\Sigma(\boldsymbol{\sigma}')\,. \tag{1}$$

That is, the prefix probability of $\boldsymbol{\sigma}$ is the sum over the probabilities of all possible continuations of $\boldsymbol{\sigma}$.

Contextual probability. The probability of a string can be factored autoregressively using prefix-conditioned probabilities. We assume that $\forall \boldsymbol{\sigma} \in \Sigma^*\colon p_\Sigma(\boldsymbol{\sigma}) > 0$.[2] First, we define the conditional prefix probability of a token $\sigma \in \Sigma$:

$$p_{\vec{\Sigma}}(\sigma \mid \boldsymbol{\sigma}) \stackrel{\text{def}}{=} \frac{p_{\vec{\Sigma}}(\boldsymbol{\sigma}\sigma)}{p_{\vec{\Sigma}}(\boldsymbol{\sigma})}. \tag{2}$$

The prefix probability of EOS is defined as

$$p_{\vec{\Sigma}}(\text{EOS} \mid \boldsymbol{\sigma}) = \frac{p_\Sigma(\boldsymbol{\sigma})}{p_{\vec{\Sigma}}(\boldsymbol{\sigma})}. \tag{3}$$

Thus, we will also use $p_{\vec{\Sigma}}(\cdot \mid \boldsymbol{\sigma})$ to denote a probability distribution over $\overline{\Sigma}$. Using the conditional prefix probabilities, there always exists a left-to-right autoregressive factorization of the probability of a string under an LM:

$$p_\Sigma(\boldsymbol{\sigma}) = p_{\vec{\Sigma}}(\text{EOS} \mid \boldsymbol{\sigma}) \prod_{t=1}^{T} p_{\vec{\Sigma}}(\sigma_t \mid \boldsymbol{\sigma}_{<t})\,. \tag{4}$$

Unigram probability. We define the unigram probability of a word $w \in \mathcal{W}$ in terms of the frequency with which it occurs in strings sampled from a language model p_Σ. Following Kiegeland et al. (2026), we assume a deterministic **unit parser** $\pi\colon \Sigma^* \to \mathcal{W}^*$ mapping token strings to word strings.[3] Its inverse, the **realization**, is a relation, since a word string is realized by many token strings. We further assume the positions of $\pi(\boldsymbol{\sigma})$ are realized by pairwise disjoint, nonempty, contiguous token spans of $\boldsymbol{\sigma}$, in left-to-right order, so that $|\pi(\boldsymbol{\sigma})| \leq |\boldsymbol{\sigma}|$. The pushforward of p_Σ through π is then defined as

$$p_{\mathcal{W}}(\boldsymbol{w}) \stackrel{\text{def}}{=} \sum_{\boldsymbol{\sigma} \in \pi^{-1}(\boldsymbol{w})} p_\Sigma(\boldsymbol{\sigma}), \qquad \boldsymbol{w} \in \mathcal{W}^*. \tag{5}$$

We write $p_{\vec{\mathcal{W}}}$ for the prefix probability of $p_{\mathcal{W}}$, so that $p_{\vec{\mathcal{W}}}(w \mid \mathbf{w})$ is the conditional next-word probability. To account for the fact that a word may admit multiple tokenizations, we write

$$\mathcal{R}(w) \stackrel{\text{def}}{=} \{\boldsymbol{\sigma} \in \Sigma^+\colon \pi(\boldsymbol{\sigma}) = w\}$$

for the set of token strings realizing w alone (identifying w with the length-one word string), where $\Sigma^+ = \Sigma^* \setminus \{\varepsilon\}$. For a token string $\boldsymbol{\sigma}$, the number of occurrences of w is read off its parse:[4]

$$\#(\boldsymbol{\sigma}, w) \stackrel{\text{def}}{=} \sum_{t=1}^{|\pi(\boldsymbol{\sigma})|} \mathbb{1}\{\pi(\boldsymbol{\sigma})_t = w\}\,. \tag{6}$$

§C describes how we implement π, and hence $\#$, for the tokenizer used in our experiments. The expected count of a word $w \in \mathcal{W}$ under p_Σ is then

$$\mu(w) \stackrel{\text{def}}{=} \mathbb{E}_{X \sim p_\Sigma}[\#(X, w)] = \sum_{\boldsymbol{\sigma} \in \Sigma^*} p_\Sigma(\boldsymbol{\sigma})\, \#(\boldsymbol{\sigma}, w)\,. \tag{7}$$

We can then derive a notion of a word's unigram probability by normalizing this expectation to a proper probability distribution over $\mathcal{W}$:

$$u(w) = \frac{\mu(w)}{\sum_{w' \in \mathcal{W}} \mu(w')}. \tag{8}$$

Note that—like the contextual prefix probability $p_{\vec{\Sigma}}(\cdot \mid \boldsymbol{\sigma})$—$u(w)$ is a distribution over $\mathcal{W}$, and is only defined for a single word $w \in \mathcal{W}$.

Remark. Let $N_{\mathcal{W}}(\boldsymbol{\sigma})$ denote the number of occurrences in $\boldsymbol{\sigma}$ of words in $\mathcal{W}$. By linearity of expectation, $\sum_{w \in \mathcal{W}} \mu(w) = \mathbb{E}_{X \sim p_\Sigma}\big[\sum_{w \in \mathcal{W}} \#(X, w)\big] = \mathbb{E}_{X \sim p_\Sigma}[N_{\mathcal{W}}(X)]$, where the last equality is immediate, since each position of $\pi(\boldsymbol{\sigma})$ is occupied by exactly one word of $\mathcal{W}$. Thus Eq. (8) normalizes by the expected word count.

3 Which Language Model?

Formal theories of language, most notably the generative tradition (Chomsky, 1957, 1965), have historically separated a speaker's idealized knowledge of grammatical rules (**competence**) from the actual, often imperfect, use of language in real-time processing (**performance**). Probability has found a central role in explaining the latter

[1] We remark that the prefix probability is not a probability distribution in general, since it is unnormalized.

[2] This holds true for transformer-based LMs due to the softmax function.

[3] Parsing into units is ambiguous in general (see Kiegeland et al. (2026)). Here, we adopt the assumption that parsing is deterministic, which holds for the units we study.

[4] A target spanning several units is treated as the corresponding word subsequence, with the count defined analogously.

(Bod et al., 2003; Jurafsky, 2003; Chater and Manning, 2006; Levy, 2008): under a probabilistic, performance-oriented view a model of language is then evaluated not by whether it generates the correct set of grammatical strings, but by how well it mirrors the behavior of human speakers. The question is thus not *whether* probability is relevant for modeling linguistic performance, but *what* distribution it should approximate. We contrast two viewpoints on this question, which we refer to as the frequentist and the cognitive views.

The frequentist view. One approach to estimating an LM is to optimize it to place high probability on a large, heterogeneous corpus—comprising, e.g., collaborative text on Wikipedia, conversational text on Reddit, and the structured, iteratively revised prose of digitized books (Radford et al., 2019; Groeneveld et al., 2024).[5] Under the frequentist view, this LM approximates the probability distribution that generated the corpus text, which we denote p_{C}. The cognitive relevance of p_{C} has been widely studied—corpus frequency has been shown to play a central role in language use and change (Bybee, 1998; Bybee and Hopper, 2001), accounting for phonological reduction (Bybee and Scheibman, 1999; Bybee, 2002) and grammaticalization (Haspelmath, 1999). On the other hand, the generative process underlying such heterogeneous corpora is complex: it involves millions of individuals writing diverse texts in an iterative fashion over extended periods of time, and the resulting corpora are several orders of magnitude larger than the linguistic input any single human observes over a lifetime (Warstadt and Bowman, 2022).

The cognitive view. In contrast, the cognitive view interprets the LM as a proxy for p_{H}—a theoretical construct representing the internalized probability distribution over linguistic strings that a human carries in their mind. Because we lack direct access to such a theoretical construct, a corpus-estimated LM often serves as a practical approximation, an assumption underlying much of psycholinguistic modeling (Wilcox et al., 2020; Oh and Schuler, 2023b; Shain et al., 2024, *inter alia*). However, as LMs have grown larger—in terms of both training data and parameter count—they have become *less* predictive of human behavior. This phenomenon is known as the inverse-scaling law (Oh and Schuler, 2023b): predictive accuracy of human data improves with scale only up to a point, after which it degrades. This motivates an alternative approach in which the LM is directly aligned with human psychometric data to obtain more cognitively plausible probability estimates (Kiegeland et al., 2024). The construct p_{H} need not be an unconstrained distribution over arbitrarily long contexts, either. The cognitive model may be *lossy* (Futrell et al., 2020), reflecting constraints such as limited working memory (Gibson, 1998, 2000) that restrict how much prior context is effectively encoded; in the limit, p_{H} reduces to unigram probabilities.

[5] Instruction-tuned (Ouyang et al., 2022) and reasoning-oriented (Guo et al., 2025) variants involve additional training objectives that shift the distribution away from raw corpus frequency, complicating a straightforwardly frequentist interpretation of the resulting probabilities.

4 Surprisal Theory

Eye tracking and reading data. To study sentence processing during reading, several corpora have been developed in which many participants read the same texts and eye-tracking data are collected while reading (Kennedy et al., 2003; Siegelman et al., 2022). Eye-tracking studies record participants' gaze continuously, allowing fixations of varying duration to be mapped directly onto the text (Rayner, 1998). Although some work has sought to model these fine-grained fixations (Re et al., 2025), the fixations are generally aggregated to yield a measure of reading time for some unit of interest. The choice of aggregation reflects assumptions about which stage of processing is being targeted (Rayner, 1998; Frank et al., 2013a; Berzak and Levy, 2023): *first fixation duration* (the duration of the reader's first fixation on a unit) is thought to reflect early lexical processing (Clifton et al., 2007), while *gaze duration* (cumulative fixation time before first leaving a unit) and *total fixation duration* (cumulative fixation time including regressions back to the unit) capture progressively later integration processes (Liversedge et al., 1998).

Contextual surprisal. Surprisal theory states that processing difficulty is related to a unit's negative log probability in a given context (Hale, 2001; Levy, 2008; Shain et al., 2024). Psycholinguistic studies test this theory by relating surprisal to behavioral or neural measures of processing effort such as fixation duration from eye-tracking data (Wilcox et al., 2023b) and event-related potentials (ERPs) from electroencephalography (Frank et al., 2013b). LMs play a central role in these studies because they provide estimates of contextual prob-

abilities. Given $p_{\mathcal{W}}$, the contextual surprisal of a unit w_t in context $\mathbf{w}_{<t}$ is defined as

$$s_{\text{context}}(w_t, \mathbf{w}_{<t}) = -\log p_{\vec{\mathcal{W}}}(w_t \mid \mathbf{w}_{<t}) \,. \quad (9)$$

Unigram surprisal. Psycholinguistic studies indicate that units with high unigram frequency are more easily accessible in the mental lexicon (Hare et al., 1995; Bybee, 1998). This is typically measured in log scale, using **unigram surprisal**

$$s_{\text{unigram}}(w) = -\log u(w) \,, \quad (10)$$

where $w \in \mathcal{W}$ is the unit of interest. Both surprisal quantities are thus functionals of the single distribution p_Σ (given the unit parser): contextual surprisal of the conditional structure it induces through $p_{\vec{\mathcal{W}}}$, and unigram surprisal of the marginal it induces through u. The consistent distribution principle (§5) demands that both be estimated from the same p_Σ, so that this relationship also holds for the estimates. Given that it has been shown to affect ease of processing, unigram surprisal is regularly included as a baseline predictor of reading times; however, some studies have focused specifically on its effect on reading (Shain, 2019, 2024; Opedal et al., 2024).

5 Principles of Frequency Estimation

Having discussed which distributions can be modeled to study language and having defined unigram and contextual surprisal, we can now formalize three guiding principles for estimating unigram surprisal and discuss relevant literature around them. The principles are summarized in Fig. 2.

Best estimate principle. The first principle adopts an empirical conception of unigram frequency. Under this conception, unigram frequency is treated as a real population quantity: the relative prevalence of a unit in the underlying distribution that generated the corpus. The observed strings are then viewed as samples from that distribution, and estimation amounts to recovering this empirical truth from data. This perspective has two advantages. First, it ties the target quantity to observable language use rather than to an unobserved latent construct. Second, it admits a simple and interpretable estimator: under independent and identically distributed (i.i.d.) samples, unigram probabilities are estimated by normalized token counts, which is equivalent to the maximum-likelihood estimator for a multinomial model; see §B. As we show in Proposition 2, because $-\log$ is strictly convex, an unbiased estimator of unigram probability does not yield an unbiased estimator of unigram surprisal. Plug-in estimators of related information-theoretic quantities (i.e., entropy) are known to be biased for the same reason (Miller, 1955; Arora et al., 2022). Even for an unbiased unigram probability estimator, the estimated unigram surprisal is biased upward. The bias in surprisal disappears only for a zero-variance unigram probability estimator, motivating lower variance frequency estimators.

Cognitive plausibility principle. The second principle adopts a cognitive conception of unigram frequency. Under this conception, the target is not an empirical population quantity directly given by attested token counts, but a latent property of the expectations in the human mind, as discussed with respect to the cognitive view of the LM (§3). The goal is not to simply recover the unigram distribution of the corpus-generating process, but to approximate the distribution that best reflects human baseline expectations over units. In the past, this has been operationalized as the extent to which frequency values are predictive of human reaction times in lexical decision tasks. This criterion has been studied as an alternative way to validate frequency resources, rather than factors like corpus size (Keuleers et al., 2010; Brysbaert et al., 2011). This has led to findings that texts from the Internet (Balota et al., 2004) or movie and TV subtitles (New et al., 2007) seem to yield more cognitively plausible frequency values than those from edited publications. Importantly, this principle may be at odds with the best estimate principle. Larger LMs, despite achieving lower perplexity, have been found to yield contextual surprisal estimates that are less predictive of processing effort than those of smaller models (Oh et al., 2022; Oh and Schuler, 2023b). One explanation for this is that larger models assign probabilities to very low-frequency units more accurately, but in ways that are less cognitively plausible (Oh et al., 2024). Similarly, Opedal et al. (2024) argue that medium-sized models may yield contextual surprisal estimates closer to unigram frequency because they are less effective at predicting rare units in context, better matching the balance of contextual and frequency-based information in human processing. The same issue arises for unigram frequency itself: the statistically best estimate need not be the cognitively most plausible one.

Although this question has been studied extensively for contextual surprisal, comparatively little is known about how modeling choices affect the cognitive plausibility of unigram frequency estimates.

Consistent distribution principle. The third principle is that contextual and unigram surprisal should be derived from the same distribution. Although the literature often estimates them from different sources, the two quantities are not independent: both are functionals of a single distribution over linguistic sequences. Contextual surprisal reflects its conditional structure, while unigram surprisal reflects the marginal distribution it induces over units. Under a shared distribution, this relationship holds by construction. Under heterogeneous estimation, it does not: the two predictors then summarize different probabilistic models rather than a single generative system. Our unified estimation framework restores this coherence by deriving both quantities from the same LM. The consistent distribution principle has been observed before, though it has primarily been addressed by changing the corpora used to calculate unigram frequency. For example, Shain (2024) calculate contextual surprisal with GPT-2. To calculate unigram surprisal, Shain use a corpus that was specifically chosen to approximate the model's training distribution in an effort to minimize effects from this mismatch. Likewise, in their assessment of surprisal as a measure of metaphor novelty, Momen and Zarrieß (2026) compare unigram surprisals from `wordfreq` (Speer, 2022) to those calculated over the training data of the LMs they use to obtain contextual surprisal.

6 From Autoregressive Language Models to Unigram Probabilities

We now consider how to operationalize the consistent distribution principle, taking an autoregressive language model as the shared distribution and studying the problem of estimating the unigram probabilities it induces. Since exact computation of Eq. (8) is infeasible, we instead consider two different estimators of this quantity. Specifically, we define estimators for words $w \in \mathcal{W}$, which may have several valid tokenizations under a language model's token alphabet Σ.[6]

[6]Even with a deterministic encoder, LMs can assign mass to non-canonical tokenizations (Vieira et al., 2025).

6.1 A Naive Monte Carlo Estimator

A straightforward way to estimate Eq. (8) is by naive Monte Carlo (NMC) sampling. Here, we independently sample N strings from p_Σ, i.e., $X^{(1)}, \ldots, X^{(N)} \overset{\text{i.i.d.}}{\sim} p_\Sigma$, and estimate $u(w)$ by the empirical frequency of w across all word occurrences in the samples:

$$\widehat{u}(w) = \frac{\sum_{n=1}^{N} \#\left(X^{(n)}, w\right)}{\sum_{n=1}^{N} \sum_{w' \in \mathcal{W}} \#\left(X^{(n)}, w'\right)}. \tag{11}$$

Proposition 1. *Assume that* $\mathbb{E}_{X \sim p_\Sigma}[|X|] < \infty$ *and that* $\sum_{w' \in \mathcal{W}} \mu(w') > 0$. *Then, for every fixed* $w \in \mathcal{W}$, *the Monte Carlo estimator* $\widehat{u}_N(w)$ *is strongly consistent and asymptotically unbiased for* $u(w)$. *That is,*

$$\widehat{u}_N(w) \xrightarrow{a.s.} u(w) \qquad \text{as } N \to \infty, \tag{12}$$

and

$$\lim_{N \to \infty} \mathbb{E}[\widehat{u}_N(w)] = u(w)\,. \tag{13}$$

6.2 A Marginalized Monte Carlo Estimator

The naive MC estimator in Eq. (11) can have high variance, since X ranges over a very large (infinite) sample space.[7] This has negative implications for the efficiency of the frequency estimator, as well as the unigram surprisal values which we derive from it. As we show in Proposition 2, unigram surprisal values are positively biased in expectation unless the estimator has zero variance, further motivating a lower-variance frequency estimator.

To reduce this variance, we propose a **marginalized Monte Carlo** (MMC) estimator. The key idea is to retain the sampled prefixes of the NMC estimator, while marginalizing over the next token at each prefix. Rather than treating a word as present or absent in the sampled next token, we compute the conditional expected contribution of that next token to the word count. Suppose a sampled sequence has reached a prefix $\boldsymbol{\sigma}$. NMC observes a single sampled continuation and therefore records either an occurrence of w at that position or no occurrence. The LM, however, assigns a probability to *every* next token, as well as to the further tokens needed to complete an occurrence of w; MMC uses all of this information.

[7]See, e.g., Giulianelli et al. (2024), who report high variance when using Monte Carlo sampling to estimate sequence-level entropy in psycholinguistic modeling. Clark et al. (2025) instead advocate for estimating entropy at the word level.

Canonical and residual occurrences. An occurrence of w may be realized by different token strings: its realizing span (§A), read as a standalone string, is an element of $\mathcal{R}(w)$ under the parser of §C. We split the count into the part realized by the canonical tokenization of w, and the part realized in any other way:

$$\#(\boldsymbol{\sigma}, w) = \#^{\text{can}}(\boldsymbol{\sigma}, w) + \#^{\text{res}}(\boldsymbol{\sigma}, w). \quad (14)$$

The canonical part can be marginalized with the LM's next-token distributions, whereas the residual part is only read off a sampled string once that string is complete.

Marginalizing at each prefix. We use $Z^{\text{can}}_{w,t}$ to refer to a canonical occurrence of w beginning at position t (formally, §A), and define

$$h_w(\sigma \mid \boldsymbol{\sigma}) \stackrel{\text{def}}{=} \mathbb{E}\left[Z^{\text{can}}_{w,t} \mid X_{<t} = \boldsymbol{\sigma},\, X_t = \sigma\right], \quad (15)$$

the probability that a canonical occurrence begun at position t is completed—including its right-boundary condition (§C)—given that the next token is σ. Averaging over the next-token distribution, $\sum_{\sigma \in \Sigma} \vec{p_\Sigma}(\sigma \mid \boldsymbol{\sigma})\, h_w(\sigma \mid \boldsymbol{\sigma})$ is the expected canonical contribution of the position following $\boldsymbol{\sigma}$.

Marginalizing the canonical part alone would discard residual occurrences. Writing $H_w(\boldsymbol{\sigma}) \stackrel{\text{def}}{=} \sum_{s=1}^{|\boldsymbol{\sigma}|} h_w(\sigma_s \mid \boldsymbol{\sigma}_{<s})$ for the marginalized canonical contribution accumulated along $\boldsymbol{\sigma}$, we define a per-step contribution that charges whatever is left over to the terminal EOS position:

$$d_w(\sigma \mid \boldsymbol{\sigma}) \stackrel{\text{def}}{=} \begin{cases} h_w(\sigma \mid \boldsymbol{\sigma}), & \sigma \in \Sigma, \\ \#(\boldsymbol{\sigma}, w) - H_w(\boldsymbol{\sigma}), & \sigma = \text{EOS}. \end{cases} \quad (16)$$

By construction, this decomposition is exact pathwise: for every completed string $\boldsymbol{\sigma}$,

$$\#(\boldsymbol{\sigma}, w) = \sum_{t=1}^{|\boldsymbol{\sigma}|} d_w(\sigma_t \mid \boldsymbol{\sigma}_{<t}) + d_w(\text{EOS} \mid \boldsymbol{\sigma}). \quad (17)$$

The EOS term therefore absorbs both the residual, non-canonical occurrences and the gap between the marginalized and the realized canonical counts.

The MMC estimator. Taking the conditional expectation of d_w at every prefix visited by a sampled string X gives the per-string count estimate

$$\widehat{\#}^{\text{MMC}}(X, w) \stackrel{\text{def}}{=} \sum_{t=1}^{|X|+1} \mathbb{E}_{A \sim \vec{p_\Sigma}(\cdot \mid X_{<t})}\left[d_w(A \mid X_{<t})\right], \quad (18)$$

which we show to be unbiased for the expected count of Eq. (7) in §A. Normalizing exactly as in Eq. (11) yields our estimator of Eq. (8):

$$\widehat{u^*}(w) = \frac{\sum_{n=1}^{N} \widehat{\#}^{\text{MMC}}\left(X^{(n)}, w\right)}{\sum_{n=1}^{N} \sum_{w' \in \mathcal{W}} \#\left(X^{(n)}, w'\right)}, \quad (19)$$

where $X^{(1)}, \ldots, X^{(N)} \overset{\text{i.i.d.}}{\sim} p_\Sigma$. The denominator is the same expected-word-count estimate used by NMC in Eq. (11).

Approximate variant. Given the computational cost of the MMC estimator (§A) and the relatively small probability mass that LMs assign to non-canonical tokenizations (Vieira et al., 2025), we also experiment with an approximation that counts only canonical occurrences, i.e., replaces $\#$ in Eq. (14) by $\#^{\text{can}}$, so that $\#^{\text{res}}(\boldsymbol{\sigma}, w) = 0$. The EOS term of the expectation-equivalent contribution $\widetilde{d}_w$ of §A then vanishes, and the corresponding estimator reduces to the canonical-only count estimate $\widehat{\#}^{\text{MMC}}_{\text{can}}(X, w) \stackrel{\text{def}}{=} \sum_{t=1}^{|X|+1} \mathbb{E}_{A \sim \vec{p_\Sigma}(\cdot \mid X_{<t})}\left[h_w(A \mid X_{<t})\right]$ (with $h_w(\text{EOS} \mid \cdot) \stackrel{\text{def}}{=} 0$), which is unbiased for the canonical share $\mathbb{E}_{X \sim p_\Sigma}[\#^{\text{can}}(X, w)]$ of Eq. (7), and so agrees with Eq. (18) in expectation exactly when the residual mass vanishes. Normalizing as before gives

$$\widehat{u^*}_{\text{can}}(w) = \frac{\sum_{n=1}^{N} \widehat{\#}^{\text{MMC}}_{\text{can}}\left(X^{(n)}, w\right)}{\sum_{n=1}^{N} \sum_{w' \in \mathcal{W}} \#\left(X^{(n)}, w'\right)}. \quad (20)$$

The approximation above does not require taking non-canonical tokenizations in the sampled strings into account, meaning that its numerator underestimates the expected count $\mu(w)$ whenever residual mass is nonzero. We study the downstream impact of this in §7.

7 Experiments

Opedal et al. (2024) model reading times of orthographic words in eleven languages using contextual surprisal from mGPT (Shliazhko et al., 2024) and unigram surprisal from the `wordfreq` package (Speer, 2022), with data from the Multilingual Eye-movement Corpus (MECO; Siegelman et al., 2022). We extend their analyses, retaining their contextual surprisal values but replacing unigram surprisal estimates obtained from `wordfreq` with those estimated from the same distribution as the

contextual surprisal using our NMC and MMC estimators. Strings for estimation are obtained by sampling from mGPT with a maximum length of 50 for computational considerations, and estimates are converted to unigram surprisals by taking the negative log of the estimated unigram probabilities.

Opedal et al. (2024) compare the predictive power of unigram surprisal, contextual surprisal, and pointwise mutual information (PMI). PMI (Fano, 1961) measures the association between a word w_t and its context $\mathbf{w}_{<t}$, and is defined as:

$$\text{PMI}(w_t, \mathbf{w}_{<t}) = s_{\text{unigram}}(w_t) - s_{\text{context}}(w_t, \mathbf{w}_{<t})\,. \tag{21}$$

Wherever PMI or unigram surprisal appear in our models, we report results using both the original `wordfreq` estimates (Speer, 2022) and each of our approaches to estimating unigram surprisal from mGPT. We model reading times using generalized additive mixed models (GAMMs), which differ from linear models by allowing predictors to enter through smooth, nonparametric functions rather than fixed linear coefficients. Our dependent variable is gaze duration. We fit five GAMMs with smooth terms for (i) unigram surprisal, (ii) contextual surprisal, (iii) PMI, (iv) contextual surprisal and unigram surprisal, and (v) PMI and unigram surprisal, each with a single spillover term per predictor and random effect smooths for participant and trial. Model fit is assessed via delta log-likelihood (Δ_{LL}), defined as the difference in per-unit log-likelihood between a candidate model and a baseline with only random effects. Predictions are obtained via 10-fold cross-validation; positive values indicate that the candidate model assigns higher likelihood to the observed reading times than this baseline. To assess whether the MMC estimators yield a different quality of fit than corpus-based unigram surprisals, we compare Δ_{LL} values between the two frequency conditions at the unit level using paired permutation tests, with p-values corrected for multiple comparisons using the Holm–Bonferroni procedure. We discuss MECO and the covered languages, preprocessing, model specification, and evaluation details in §F.[8]

[8]We present more configurations of results (e.g., under linear models, different reading time measures and corpora, and different preprocessing decisions) at https://github.com/replicaverse/replica/tree/main/src/replica/studies/hopton2026unigram.

J	MMC variance	NMC variance	Ratio
100	1.685e-08	1.736e-07	10.30
200	8.508e-09	9.689e-08	11.39
300	5.648e-09	6.769e-08	11.98
400	4.044e-09	5.448e-08	13.47
500	3.307e-09	4.281e-08	12.94

Table 1: Mean bootstrap variance of the Marginalized (MMC) and Naive (NMC) Monte Carlo unigram frequency estimators across subsample sizes J. Ratio = NMC / MMC.

8 Results

8.1 Marginalizing Reduces Variance

To study empirically whether the variance of the MMC estimator is lower than that of the naive Monte Carlo estimator in Eq. (11), we use both to compute estimates of unigram frequency for each unigram appearing in the English portion of MECO (Siegelman et al., 2022). Given $N = 500$ samples, we assess the variance of an estimator $\widehat{\mu}$ via bootstrapping at subsample sizes $J \in \{100, 200, 300, 400, 500\}$, drawing $K = 1000$ resamples with replacement and computing the estimator on each resample. The variance is then estimated as follows:

$$\begin{aligned}\mathbb{V}(\widehat{\mu}) &= \mathbb{E}\left[(\widehat{\mu} - \mathbb{E}[\widehat{\mu}])^2\right] \\ &= \frac{1}{K-1}\sum_{k=1}^{K}\left(\widehat{\mu_k} - \overline{\widehat{\mu}}\right)^2,\end{aligned} \tag{22}$$

where $\overline{\widehat{\mu}} \overset{\text{def}}{=} \frac{1}{K}\sum_{k=1}^{K}\widehat{\mu}_k$. Results are in Tab. 1 (see Fig. 4 for visualizations). The marginalized estimator (Eq. (19)) exhibits consistently lower variance than the naive estimator (Eq. (11)) across all subsample sizes and unigrams. Indeed, the ratio of the two variances generally grows with larger subsamples. This confirms that marginalization is an empirically efficient method to estimate unigram probabilities.

We also investigate the coverage of the naive Monte Carlo estimate, i.e., for how many unigrams in MECO it successfully obtains estimates. Obtaining an estimate with the Naive Monte Carlo estimator requires the unigram to occur in the sample pool, which is not guaranteed. Across a range of mGPT sample sizes from $N = 10^2$ to $N = 10^7$, we find that NMC never successfully obtains estimates for 100% of the types in the MECO corpus for any of the 11 languages tested here, and even obtains less than 50% coverage for Finnish types in sample pools of 10^7 strings (see §E).

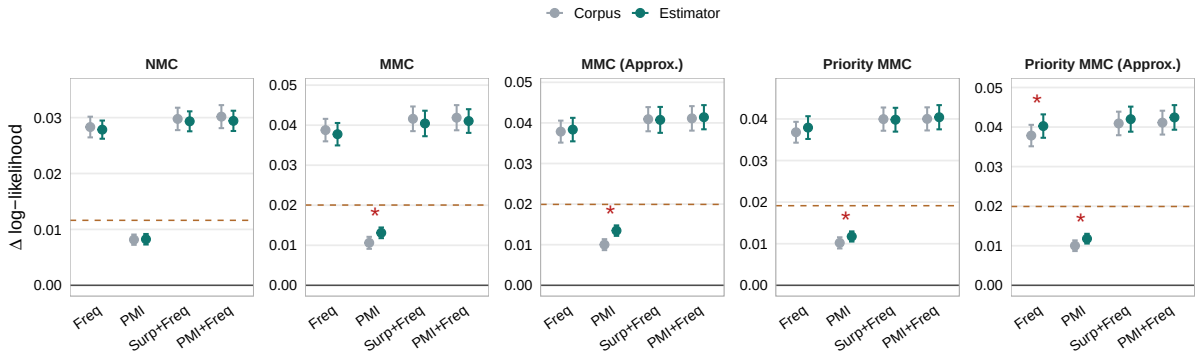


Figure 3: Comparison of Δ_{LL} for various GAMMs of orthographic word gaze duration fit with unigram surprisals obtained with the `wordfreq` package (Speer, 2022) or with one of our estimators using samples from mGPT. Values are averaged across languages. Dashed lines represent the Δ_{LL} of adding contextual surprisal relative to each facet's baseline for reference. Significance tests compare estimated unigram surprisal against the corpus-estimated values via paired permutation test on item-level Δ_{LL} differences, with stars marking corrected significance at an α level of 0.05. Error bars represent 95% confidence intervals.

Importantly, our coverage experiment indicates that sampled text from mGPT is skewed toward Hebrew, Greek, and Russian. Thus, while the MMC estimator does not have the same coverage problems as the NMC estimator, the sampled contexts used for estimates may not be informative (i.e., because they are from different languages). To better adapt our estimator for use with multilingual language models, we introduce priority-MMC, which draws samples from a large pool with language-specific strata (see §I for details).

8.2 Downstream Impact of Marginalizing

We now focus on models of gaze duration fit with various unigram estimates: corpus-based estimates from `wordfreq` (Speer, 2022), NMC estimates, MMC estimates and their canonical-only approximations (MMC and MMC Approx.), and priority-MMC estimates and their canonical-only approximations (Priority MMC and Priority MMC Approx.). We present bootstrapped distributions of item-level Δ_{LL} for GAMMs modeling gaze duration in Fig. 3. Across all MMC variants, we consistently observe a significant increase in Δ_{LL} for GAMMs fit with PMI calculated with MMC unigram surprisals, relative to GAMMs fit with PMI calculated with `wordfreq` unigram surprisals. As in prior work (Opedal et al., 2024; Heilbron et al., 2023), we find that unigram surprisal is a stronger predictor of reading times than contextual surprisal; our results confirm this finding in an environment where both values come from the same underlying distribution. We note that because of relatively poor coverage (even at high sample sizes) for rarer words, more samples are excluded from the NMC models, meaning the results are likely less robust than those from the various MMC estimators. Though the MMC and priority MMC estimators properly account for non-canonical tokenizations, our findings indicate that approximations over only canonical samples yield similar results with higher coverage and less computational cost. The results highlight that principled approaches to estimating unigram surprisal have a downstream impact.

9 Conclusion

This paper highlights an understudied matter in surprisal theory literature—while much attention has been given to how we estimate contextual surprisal, little research has focused on how we estimate unigram surprisal. We propose three principles to guide this estimation and introduce a Monte Carlo estimator of unigram surprisal from a language model, along with several marginalized variants. By deriving both surprisal quantities from the same model, differences in predictive power can be attributed to modeling decisions rather than mismatched estimation procedures. Our experiments show that the choice of unigram surprisal estimator has empirical consequences, and we provide further support that contextual surprisal is a weaker predictor of reading times than unigram surprisal in modeling reading time. We hope this encourages future work on what modeling decisions result in cognitively plausible unigram surprisal estimations and a more careful consideration of the role of frequency in computational psycholinguistics.

Limitations

Though we argue that our principles for estimating unigram surprisal are applicable to computational psycholinguistics as a whole, we limit our analyses to sentence processing. We find that the procedure for estimating unigram surprisal has an empirical impact when it comes to modeling reading times, but the claim warrants further investigation in other lines of computational psycholinguistics research, e.g., in modeling speech segmentation during language acquisition or acceptability judgment experiments.

In general, estimating unigram surprisal from a language model is computationally expensive compared to using lexical databases or corpus counts, particularly for the marginalized estimators we present. Because of their lower variance, the marginalized variants of our estimator also address this limitation of our method to some extent: relatively few samples are required to obtain stable estimates compared to the Naive Monte Carlo estimator. Still, computational resources limited our sample size and the maximum string length of the strings sampled from mGPT.

We also study a limited scope of units. Our implementation of the unit parser matches whitespace-initial token spans (§C), which is adequate for the whitespace-delimited orthographies studied here but not in general. Whether the impact of the specific estimator used varies with units of interest (for example, characters or the contextually defined words studied in Kiegeland et al. (2026)) remains an open question.

Ethical Concerns

This study uses eye-tracking data from human participants. Because the dataset has already been published (Siegelman et al., 2022), we foresee no new ethical concerns from the contributions made in this paper.

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A Estimator description

- **Word occurrences.** Occurrences of a word w and the split of $\#$ into canonical and residual realizations are as in Eq. (14), writing $C_w(X) \stackrel{\text{def}}{=} \#(X, w)$ and likewise for the canonical and residual parts: an occurrence at unit position s of $\pi(X)$ is *canonical* when the token span realizing it equals the canonical tokenization of w, and *residual* otherwise. We write $Z^{\text{can}}_{w,t}$ for the indicator that a canonical occurrence begins at token position t.

- **Marginalize the canonical realization.** A canonical occurrence of w starting at position t requires the next k tokens to match its canonical tokenization, followed by a valid right boundary. We therefore define

$$h_w(a \mid x) \stackrel{\text{def}}{=} \mathbb{E}\big[Z^{\text{can}}_{w,t} \mid X_{<t} = x,\, X_t = a\big]\,.$$

  We can then define the probability that the canonical realization of w begins at prefix x and is completed with a valid right boundary as

$$\sum_{a \in \Sigma} \vec{p_\Sigma}(a \mid x)\, h_w(a \mid x).$$

  Specifically, computing this quantity at a prefix x requires $O((k+1)C)$ computation, where k is the number of tokens in the canonical realization of w and C is the cost of a single LM forward pass.

- **Recover the exact count at EOS.** Define the per-step contribution

$$d_w(a \mid x) = \begin{cases} h_w(a \mid x), & a \in \Sigma, \\ C^{\text{can}}_w(x) + C^{\text{res}}_w(x) - \displaystyle\sum_{s=1}^{|x|} h_w(x_s \mid x_{<s}), & a = \text{EOS}. \end{cases}$$

  Then, for every completed string X,

$$\begin{aligned} C_w(X) &= C^{\text{can}}_w(X) + C^{\text{res}}_w(X) \\ &= \sum_{t=1}^{|X|} h_w(X_t \mid X_{<t}) + \left[C^{\text{can}}_w(X) + C^{\text{res}}_w(X) - \sum_{t=1}^{|X|} h_w(X_t \mid X_{<t}) \right] \\ &= \sum_{t=1}^{|X|} d_w(X_t \mid X_{<t}) + d_w(\text{EOS} \mid X). \end{aligned}$$

- **Marginalize the per-step contribution d_w.** At each prefix x, take the conditional expectation of the next-step contribution:

$$\mathbb{E}_{A \sim \vec{p_\Sigma}(\cdot \mid x)}\left[d_w(A \mid x)\right] = \sum_{a \in \Sigma} \vec{p_\Sigma}(a \mid x)\, h_w(a \mid x) + \vec{p_\Sigma}(\text{EOS} \mid x) \left[C^{\text{can}}_w(x) + C^{\text{res}}_w(x) - \sum_{s=1}^{|x|} h_w(x_s \mid x_{<s}) \right].$$

- **Define the marginalized estimator.** For a sampled string X, sum the conditional expected contributions over all visited prefixes:

$$\widehat{C}^{\text{MMC}}_w(X) = \sum_{t=1}^{|X|+1} \mathbb{E}_{A \sim \vec{p_\Sigma}(\cdot \mid X_{<t})}\left[d_w(A \mid X_{<t})\right].$$

- **Show unbiasedness.** Let $\tau(X) \stackrel{\text{def}}{=} |X| + 1$. Then

$$
\begin{aligned}
\mathbb{E}_{X\sim p}\left[\widehat{C}_w^{\mathrm{MMC}}(X)\right] &= \mathbb{E}_{X\sim p}\left[\sum_{t=1}^{\tau(X)} \mathbb{E}_{A\sim \vec{p_\Sigma}(\cdot|X_{<t})}\left[d_w(A \mid X_{<t})\right]\right] \\
&\textit{(extend sum via indicator)} \quad = \mathbb{E}_{X\sim p}\left[\sum_{t=1}^{\infty} \mathbf{1}\{|X| \geq t-1\}\mathbb{E}_{A\sim \vec{p_\Sigma}(\cdot|X_{<t})}\left[d_w(A \mid X_{<t})\right]\right] \\
&\textit{(expectation as sum)} \quad = \sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x}) \sum_{t=1}^{\infty} \mathbf{1}\{|\boldsymbol{x}| \geq t-1\}\mathbb{E}_{A\sim \vec{p_\Sigma}(\cdot|\boldsymbol{x}_{<t})}\left[d_w(A \mid \boldsymbol{x}_{<t})\right] \\
&\textit{(swap sums (Fubini))} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x})\mathbf{1}\{|\boldsymbol{x}| \geq t-1\}\mathbb{E}_{A\sim \vec{p_\Sigma}(\cdot|\boldsymbol{x}_{<t})}\left[d_w(A \mid \boldsymbol{x}_{<t})\right] \\
&\textit{(rewrite as expectation)} \quad = \sum_{t=1}^{\infty} \mathbb{E}_{X\sim p}\left[\mathbf{1}\{|X| \geq t-1\}\mathbb{E}_{A\sim \vec{p_\Sigma}(\cdot|X_{<t})}\left[d_w(A \mid X_{<t})\right]\right] \\
&\textit{(expectation as sum)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x})\, \mathbf{1}\{|\boldsymbol{x}| \geq t-1\}\mathbb{E}_{A_t\sim \vec{p_\Sigma}(\cdot|\boldsymbol{x}_{<t})}\left[d_w(A_t \mid \boldsymbol{x}_{<t})\right] \\
&\textit{(split } \boldsymbol{x} = \boldsymbol{x}_{<t}\boldsymbol{z}\textit{)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}_{<t}\in\Sigma^{t-1}} \sum_{\boldsymbol{z}\in\Sigma^*} \left[\prod_{s=1}^{t-1} \vec{p_\Sigma}(x_s \mid \boldsymbol{x}_{<s})\right] \left[\prod_{r=1}^{|\boldsymbol{z}|} \vec{p_\Sigma}(z_r \mid \boldsymbol{x}_{<t}\boldsymbol{z}_{<r})\right] \\
&\qquad \times \vec{p_\Sigma}(\text{EOS} \mid \boldsymbol{x}_{<t}\boldsymbol{z})\mathbf{1}\{|\boldsymbol{x}_{<t}\boldsymbol{z}| \geq t-1\}\mathbb{E}_{A_t\sim \vec{p_\Sigma}(\cdot|\boldsymbol{x}_{<t})}\left[d_w(A_t \mid \boldsymbol{x}_{<t})\right] \\
&\textit{(}\boldsymbol{z}\textit{-sum equals 1)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}_{<t}\in\Sigma^{t-1}} \underbrace{\left[\prod_{s=1}^{t-1} \vec{p_\Sigma}(x_s \mid \boldsymbol{x}_{<s})\right]}_{=\vec{p_\Sigma}(\boldsymbol{x}_{<t})} \\
&\qquad \times \underbrace{\sum_{\boldsymbol{z}\in\Sigma^*} \left[\prod_{r=1}^{|\boldsymbol{z}|} \vec{p_\Sigma}(z_r \mid \boldsymbol{x}_{<t}\boldsymbol{z}_{<r})\right] \vec{p_\Sigma}(\text{EOS} \mid \boldsymbol{x}_{<t}\boldsymbol{z})}_{=1} \\
&\qquad \times \mathbf{1}\{|\boldsymbol{x}_{<t}\boldsymbol{z}| \geq t-1\}\mathbb{E}_{A_t\sim \vec{p_\Sigma}(\cdot|\boldsymbol{x}_{<t})}\left[d_w(A_t \mid \boldsymbol{x}_{<t})\right] \\
&\textit{(indicator equals 1)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}_{<t}\in\Sigma^{t-1}} \vec{p_\Sigma}(\boldsymbol{x}_{<t}) \underbrace{\mathbf{1}\{|\boldsymbol{x}_{<t}| \geq t-1\}}_{=1 \text{ since } \boldsymbol{x}_{<t}\in\Sigma^{t-1}} \mathbb{E}_{A_t\sim \vec{p_\Sigma}(\cdot|\boldsymbol{x}_{<t})}\left[d_w(A_t \mid \boldsymbol{x}_{<t})\right] \\
&\textit{(expand inner expectation)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}_{<t}\in\Sigma^{t-1}} \vec{p_\Sigma}(\boldsymbol{x}_{<t}) \sum_{a\in\Sigma\cup\{\text{EOS}\}} \vec{p_\Sigma}(a \mid \boldsymbol{x}_{<t})\, d_w(a \mid \boldsymbol{x}_{<t}) \\
&\textit{(split off EOS term)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}_{<t}\in\Sigma^{t-1}} \vec{p_\Sigma}(\boldsymbol{x}_{<t}) \Big[\sum_{a\in\Sigma} \vec{p_\Sigma}(a \mid \boldsymbol{x}_{<t}) d_w(a \mid \boldsymbol{x}_{<t}) \\
&\qquad\qquad + \vec{p_\Sigma}(\text{EOS} \mid \boldsymbol{x}_{<t}) d_w(\text{EOS} \mid \boldsymbol{x}_{<t})\Big] \\
&\textit{(chain rule of } \vec{p_\Sigma}\textit{)} \quad = \sum_{t=1}^{\infty} \sum_{\boldsymbol{x}_{\leq t}\in\Sigma^{t}} \vec{p_\Sigma}(\boldsymbol{x}_{\leq t}) d_w(x_t \mid \boldsymbol{x}_{<t})
\end{aligned}
$$

$$+\sum_{t=1}^{\infty}\sum_{\boldsymbol{x}_{<t}\in\Sigma^{t-1}} p_{\vec{\Sigma}}(\boldsymbol{x}_{<t})p_{\vec{\Sigma}}(\text{EOS}\mid \boldsymbol{x}_{<t})d_w(\text{EOS}\mid \boldsymbol{x}_{<t})$$

(EOS term over full strings)
$$=\sum_{t=1}^{\infty}\sum_{\boldsymbol{x}_{\leq t}\in\Sigma^{t}} p_{\vec{\Sigma}}(\boldsymbol{x}_{\leq t})d_w(x_t\mid \boldsymbol{x}_{<t})+\sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x})d_w(\text{EOS}\mid \boldsymbol{x})$$

$$=\sum_{t=1}^{\infty}\sum_{\boldsymbol{x}_{\leq t}\in\Sigma^{t}}\sum_{\boldsymbol{z}\in\Sigma^*} p(\boldsymbol{x}_{\leq t}\boldsymbol{z})d_w(x_t\mid \boldsymbol{x}_{<t})+\sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x})d_w(\text{EOS}\mid \boldsymbol{x})$$

(regroup by $\boldsymbol{x}$)
$$=\sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x})\sum_{t=1}^{|\boldsymbol{x}|} d_w(x_t\mid \boldsymbol{x}_{<t})+\sum_{\boldsymbol{x}\in\Sigma^*} p(\boldsymbol{x})d_w(\text{EOS}\mid \boldsymbol{x})$$

(definition of C_w)
$$=\mathbb{E}_{X\sim p}\left[\sum_{t=1}^{|X|} d_w(X_t\mid X_{<t})+d_w(\text{EOS}\mid X)\right]=\mathbb{E}_{X\sim p}\left[C_w(X)\right].$$

- **Define an expectation-equivalent contribution.** Because the canonical correction term has zero expectation,

$$\mathbb{E}_{X\sim p}\left[C_w^{\text{can}}(X)-\sum_{t=1}^{|X|} h_w(X_t\mid X_{<t})\right]=0,$$

we can remove it from the EOS contribution without changing the expected count. We therefore define

$$\widetilde{d}_w(a\mid x)\stackrel{\text{def}}{=}\begin{cases} h_w(a\mid x), & a\in\Sigma,\\ C_w^{\text{res}}(x), & a=\text{EOS}.\end{cases}$$

The corresponding cumulative quantity is

$$\widetilde{C}_w(X)\stackrel{\text{def}}{=}\sum_{t=1}^{|X|}\widetilde{d}_w(X_t\mid X_{<t})+\widetilde{d}_w(\text{EOS}\mid X)=\sum_{t=1}^{|X|} h_w(X_t\mid X_{<t})+C_w^{\text{res}}(X).$$

Unlike d_w, the contribution $\widetilde{d}_w$ does not recover the exact count pathwise. However,

$$\begin{aligned}\mathbb{E}_{X\sim p}\left[\widetilde{C}_w(X)\right]&=\mathbb{E}_{X\sim p}\left[\sum_{t=1}^{|X|} h_w(X_t\mid X_{<t})+C_w^{\text{res}}(X)\right]\\ &=\mathbb{E}_{X\sim p}\left[C_w^{\text{can}}(X)+C_w^{\text{res}}(X)\right]\\ &=\mathbb{E}_{X\sim p}\left[C_w(X)\right].\end{aligned}$$

Thus, $\widetilde{d}_w$ yields an expectation-equivalent decomposition of the count, whereas d_w yields a pathwise-exact decomposition.

- **Define the expectation-equivalent marginalized estimator.**

Using $\widetilde{d}_w$, we define the corresponding marginalized estimator as

$$\widehat{C}_w^{\text{MMC-}\widetilde{d}}(X)\stackrel{\text{def}}{=}\sum_{t=1}^{|X|+1}\mathbb{E}_{A\sim p_{\vec{\Sigma}}(\cdot\mid X_{<t})}\left[\widetilde{d}_w(A\mid X_{<t})\right].$$

B MLE of Unigram Frequency

Suppose we observe strings $X^{(1)}, \ldots, X^{(N)} \in \Sigma^*$ sampled from some distribution over strings. A direct estimator of unigram probability is obtained by replacing the expectations in Eqs. (7) and (8) with empirical averages.

First, define the empirical expected count

$$\hat{\mu}_N(w) \stackrel{\text{def}}{=} \frac{1}{N} \sum_{n=1}^{N} \#\left(X^{(n)}, w\right). \tag{23}$$

Plugging this and its sum over $\mathcal{W}$ into Eq. (8) yields the estimator

$$\hat{p}_N(w) \stackrel{\text{def}}{=} \frac{\hat{\mu}_N(w)}{\sum_{w' \in \mathcal{W}} \hat{\mu}_N(w')}. \tag{24}$$

Equivalently, $\hat{p}_N(w)$ is just the relative frequency of w among all word occurrences in the sample. The corresponding plug-in estimator of unigram surprisal is

$$\hat{s}_{\text{uni},N}(w) \stackrel{\text{def}}{=} -\log \hat{p}_N(w). \tag{25}$$

This is the standard corpus-based approach to unigram estimation. It is a *plug-in* estimator because it estimates the population moments appearing in the definition of unigram probability and then substitutes those estimates into the target functional. Lexical resources such as frequency tables or unigram databases can be viewed as implementations of this same basic strategy, possibly with additional preprocessing or smoothing.

Proposition 2. *For an unbiased estimator of unigram probability, the expected surprisal of the frequency estimator overestimates the true unigram surprisal:*

$$\mathbb{E}[-\log \hat{p}(w)] \;\geq\; -\log u(w) \;=\; s_{\textit{unigram}}(w). \tag{26}$$

Equality holds if and only if $\hat{p}(w)$ has zero variance, i.e., is almost surely constant.

Proof. The expected surprisal of the frequency estimator is $\mathbb{E}\left[-\log \hat{p}\,(w)\right]$. Since $-\log$ is strictly convex,

$$\mathbb{E}\left[-\log \hat{p}\,(w)\right] \overset{\text{Jensen}}{\geq} -\log \mathbb{E}\left[\hat{p}\,(w)\right] \overset{\text{unbiased}}{=} -\log u\,(w) = s_{\text{unigram}}\,(w), \tag{27}$$

where the last equality uses $\mathbb{E}\left[\hat{p}\,(w)\right] = u\,(w)$. Since $-\log$ is *strictly* convex, the Jensen step is strict unless $\hat{p}\,(w)$ equals $\mathbb{E}\left[\hat{p}\,(w)\right]$ almost surely. That is, unless it has zero variance, in which case the bias disappears. □

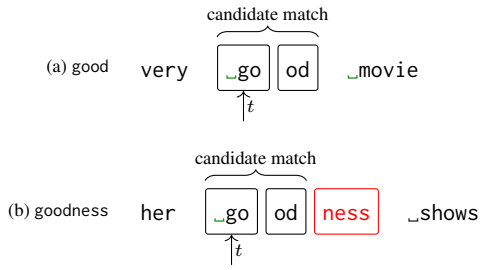
C Implementing the Unit Parser

In practice, we do not compute π exactly, but approximate it, and hence the count $\#$, by matching whitespace-initial token spans. Kiegeland et al. (2026) argue that whitespace heuristics of this kind are not linguistically adequate in general. We adopt the heuristic as an implementation of π for the whitespace-delimited words studied here and note this as a limitation.

In the following example, we illustrate our boundary-sensitive matching for the target word `good`. ␣ marks the leading whitespace carried by a token, and green ␣ symbols indicate the word boundaries relevant to the match.

In both cases the scan starts at token position t on `␣go`, whose whitespace provides the left boundary. In (a), the span `␣go od` forms `good` and the whitespace of the next token `␣movie` provides a valid right boundary, so the occurrence is counted. In (b), the same span is followed by `ness`, which carries no whitespace, so the right boundary condition fails and the occurrence is not counted.

Below we enumerate the left and right boundaries the matching requires:



Boundary	Location	Condition (satisfied when)
Left	first token of w	it carries a leading ␣ (word-initial marker Ġ), *or* it begins the sequence
Right	token following the last token of w	it carries a leading ␣ (marker Ġ), *or* it is EOS

D Properties of the Naive Monte Carlo Estimator

Proposition 1. *Assume that* $\mathbb{E}_{X\sim p_\Sigma}[|X|] < \infty$ *and that* $\sum_{w'\in\mathcal{W}} \mu(w') > 0$. *Then, for every fixed* $w \in \mathcal{W}$, *the Monte Carlo estimator* $\widehat{u}_N(w)$ *is strongly consistent and asymptotically unbiased for* $u(w)$. *That is,*

$$\widehat{u}_N(w) \xrightarrow{a.s.} u(w) \qquad \text{as } N \to \infty, \tag{28}$$

and

$$\lim_{N\to\infty} \mathbb{E}[\widehat{u}_N(w)] = u(w)\,. \tag{29}$$

Proof. Recall that $N_{\mathcal{W}}(\boldsymbol{\sigma}) = \sum_{w'\in\mathcal{W}} \#(\boldsymbol{\sigma}, w')$ denotes the number of occurrences in $\boldsymbol{\sigma}$ of words in $\mathcal{W}$, so that the estimator in Eq. (11) can be written as

$$\widehat{u}_N(w) = \frac{\sum_{n=1}^{N} \#\left(X^{(n)}, w\right)}{\sum_{n=1}^{N} N_{\mathcal{W}}\left(X^{(n)}\right)},$$

with the convention $\widehat{u}_N(w) \stackrel{\text{def}}{=} 0$ when the denominator is zero. For any $N > 0$, define the random vector

$$Z_N = \left(\frac{1}{N}\sum_{n=1}^{N} \#\left(X^{(n)}, w\right), \frac{1}{N}\sum_{n=1}^{N} N_{\mathcal{W}}\left(X^{(n)}\right)\right).$$

Further define

$$\ell = \mathbb{E}_{X\sim p_\Sigma}[N_{\mathcal{W}}(X)] = \sum_{w'\in\mathcal{W}} \mu(w')\,.$$

Since every occurrence of a word occupies at least one token position,

$$0 \le \#(X, w) \le N_{\mathcal{W}}(X) \le |X|,$$

the assumption $\mathbb{E}_{X\sim p_\Sigma}[|X|] < \infty$ implies that

$$\mathbb{E}_{X\sim p_\Sigma}[\#(X, w)] < \infty \qquad \text{and} \qquad \mathbb{E}_{X\sim p_\Sigma}[N_{\mathcal{W}}(X)] < \infty.$$

Hence the strong law of large numbers applies to both coordinates of Z_N, and

$$Z_N \xrightarrow{a.s.} (\mu(w)\,, \ell)\,. \tag{30}$$

Now define $g\colon \mathbb{R} \times (\mathbb{R}\setminus\{0\}) \to \mathbb{R}, \quad g(x, y) = \frac{x}{y}$. Since g is continuous at every point (x, y) with $y \neq 0$, and since $\ell > 0$ by assumption (so that, almost surely, the denominator is nonzero for all sufficiently large N), we can apply the continuous mapping theorem (Van der Vaart, 2000, Theorem 2.3)

$$\widehat{u}_N(w) = g\left(\frac{1}{N}\sum_{n=1}^{N} \#\left(X^{(n)}, w\right), \frac{1}{N}\sum_{n=1}^{N} N_{\mathcal{W}}\left(X^{(n)}\right)\right) \xrightarrow{a.s.} g(\mu(w)\,, \ell) = \frac{\mu(w)}{\ell} = u(w)\,.$$

This proves strong consistency. It remains to show asymptotic unbiasedness. Since $\widehat{u}_N(w) \to u(w)$ almost surely and, because $\#(X, w) \leq N_{\mathcal{W}}(X)$, we have $0 \leq \widehat{u}_N(w) \leq 1$ for every $N > 0$, by the bounded convergence theorem (Durrett, 2019, Theorem 1.6.7)

$$\lim_{N\to\infty} \mathbb{E}[\widehat{u}_N(w)] = \mathbb{E}\left[\lim_{N\to\infty} \widehat{u}_N(w)\right] = u(w). \tag{31}$$

Thus

$$\lim_{N\to\infty} (\mathbb{E}[\widehat{u}_N(w)] - u(w)) = 0, \tag{32}$$

and the estimator is asymptotically unbiased. □

E Supplementary Estimator Results

E.1 Variance Analysis

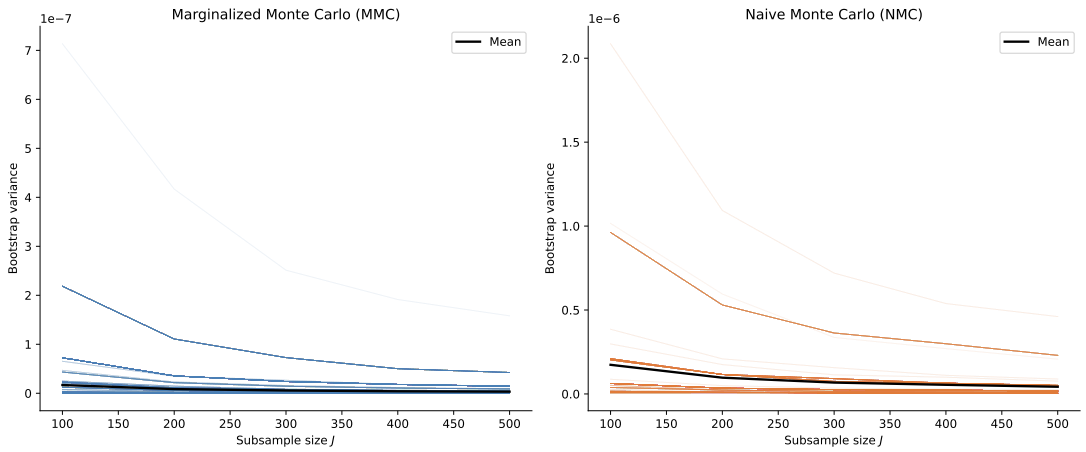


Figure 4: Visualization of the bootstrapped variance estimates of our marginalized and naive Monte Carlo estimators of unigram frequency. Each line represents a single unigram from the English MECO data. The "Mean" lines refer to the average variance across all unigrams.

E.2 Naive Monte Carlo Coverage

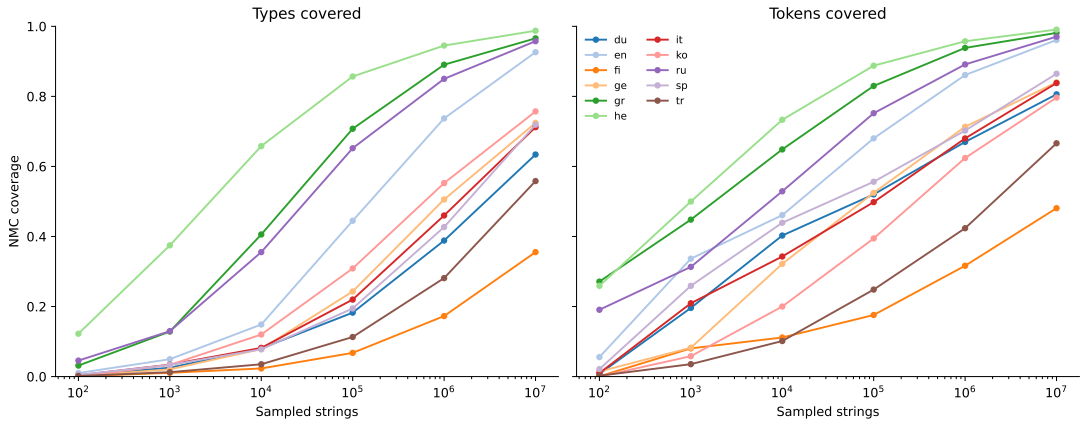


Figure 5: Proportion of types and tokens in the MECO corpus for which estimates can be obtained from our Naive Monte Carlo estimator at a range of sample sizes. Samples are obtained from mGPT with the maximum length hyperparameter set to 50 tokens. Estimates cannot be obtained when the target type/token never appears in the N sampled strings.

E.3 Runtime Analysis

We choose five unigrams each of lengths 1–4 tokens from the English MECO dataset and estimate their unigram probability five times each with the naive MC algorithm in Eq. (11) and the marginalized variant in Eq. (19). Experiments are run on an NVIDIA H100 NVL GPU.

Our results demonstrate that for unigrams of token length $T = 1$, both estimators incur similar costs, which are mainly from the generation of $N = 500$ strings. For multi-token unigrams, however, the MMC estimator scales with token count while times for the NMC estimator are stable. This is a consequence of our estimation of the joint probability of all the tokens in multi-token unigrams, which requires $T - 1$ additional rounds of batched forward passes over all positions in the sampled contexts. Still, this cost is paid only once per unit in the dataset, and the faster NMC estimates for multi-token words come at the cost of higher variance estimates relative to the MMC estimator (see Tab. 1 and Fig. 4).

Tokens	NMC (s)	MMC (s)
1	3.788 (0.003)	86.269 (0.101)
2	3.788 (0.003)	166.344 (0.161)
3	3.787 (0.002)	247.479 (0.105)
4	3.783 (0.008)	329.654 (0.153)

Table 2: Mean (SD) wall-clock time per estimate for the Naive (NMC) and Marginalized (MMC) Monte Carlo estimators, broken down by the number of subword tokens in the target unigram ($N = 500$ samples, batch size 128).

F Experiment Details

F.1 Reading Time Data

Language	Participants	Word Types	Observations	Train Size	Test Size
Dutch	45	662	31680	28512	3168
English	46	930	55120	49608	5512
Finnish	49	306	14853	13368	1485
German	45	753	43942	39548	4394
Greek	45	945	39972	35975	3997
Hebrew	47	1303	49430	44487	4943
Italian	54	791	48097	43287	4810
Korean	32	916	15364	13828	1536
Spanish	48	831	48448	43603	4845
Turkish	29	528	12083	10875	1208
Russian	46	1158	46641	41977	4664

Table 3: Per-language counts for the NMC estimator: observations entering the gaze-duration model are per participant. Train/test split refer to the size of the folds used to fit and test each of the GAMMs.

Language	Participants	Word Types	Observations	Train Size	Test Size
Dutch	45	996	44816	40334	4482
English	46	945	55792	50213	5579
Finnish	49	1042	49783	44805	4978
German	45	1037	58171	52354	5817
Greek	45	845	35558	32002	3556
Hebrew	47	954	36248	32623	3625
Italian	54	1051	61687	55518	6169
Korean	32	1208	21057	18951	2106
Spanish	48	1036	58185	52367	5818
Turkish	29	1032	22935	20641	2294
Russian	46	1133	45621	41059	4562

Table 4: Per-language counts for the MMC estimator: observations entering the gaze-duration model are per participant. Train/test split refer to the size of the folds used to fit and test each of the GAMMs.

All reading time data used in our models is obtained from the L1 portion of the MECO corpus (Siegelman et al., 2022). Our selection of the eleven languages used here represents the overlap between languages in MECO and those supported by the LM used in our experiments, mGPT. In preprocessing, we remove words skipped by participants and obtain first fixation duration (i.e., the length of the participant's first fixation on a word), gaze duration (the total time a participant fixates on a word before leaving it the first time) and the total fixation duration (the total time a participant fixates on a word regardless of whether they left the word and came back to it later). In their description of MECO, Siegelman et al. (2022) report on their procedure for aligning fixations to words and filtering data due to poor eye-tracker calibration. Tables throughout the section give the fold sizes in the 10-fold cross-validation used for GAMM evaluation (see §7), which differ between estimators due to differences in coverage.

Language	Participants	Word Types	Observations	Train Size	Test Size
Dutch	45	1009	45314	40783	4531
English	46	980	57883	52095	5788
Finnish	49	1076	51962	46766	5196
German	45	1047	58861	52975	5886
Greek	45	975	41229	37106	4123
Hebrew	47	1319	50126	45113	5013
Italian	54	1071	62485	56237	6248
Korean	32	1251	21739	19565	2174
Spanish	48	1045	58554	52699	5855
Turkish	29	1056	23440	21096	2344
Russian	46	1194	48066	43259	4807

Table 5: Per-language counts for the MMC (Approx.) estimator: observations entering the gaze-duration model are per participant. Train/test split refer to the size of the folds used to fit and test each of the GAMMs.

Language	Participants	Word Types	Observations	Train Size	Test Size
Dutch	45	907	41247	37122	4125
English	46	965	56924	51232	5692
Finnish	49	823	39298	35368	3930
German	45	950	53534	48181	5353
Greek	45	957	40455	36409	4046
Hebrew	47	1303	49454	44509	4945
Italian	54	995	58600	52740	5860
Korean	32	1118	19156	17240	1916
Spanish	48	994	56012	50411	5601
Turkish	29	894	20149	18134	2015
Russian	46	1176	47253	42528	4725

Table 6: Per-language counts for the Priority MMC estimator: observations entering the gaze-duration model are per participant. Train/test split refer to the size of the folds used to fit and test each of the GAMMs.

mGPT is a 1.3B parameter decoder-only language model pretrained on data from 61 languages. Shliazhko et al. (2024) report using Wikipedia documents and the C4 dataset to pretrain mGPT. `wordfreq` estimates are fit with data from Wikipedia, subtitle and news corpora, books, web crawls, and social media. While there it is feasible that there is some overlap in the domain and content of the two resources' data sources, they are not fit on identical data.

F.2 Model Specification and Evaluation

As an example of how our GAMMs of reading times are structured, the formula for our surprisal-only model is as follows:

$$\begin{aligned} \text{RT}_i = \beta_0 + f_1(s_{\text{context},i}) + f_2(s_{\text{context},i-1}) \\ + b_{\text{participant}(i)} + b_{\text{trial}(i)} + \epsilon_i. \end{aligned} \tag{33}$$

Above, f_1 and f_2 are main-effect smooths of the surprisal of the current and previous unit, respectively. $b_{\text{participant}(i)}$ and $b_{\text{trial}(i)}$ represent random effects that account for participant and trial variation. We specifically implement main-effect smooths as cubic regression splines limited to six basis functions to avoid overfitting, as in Opedal et al. (2024). ϵ_i represents a Gaussian noise term. Predictors are standardized prior to model fitting, and we fit all GAMMs using the `bam()` function of the `mgcv` package (Wood, 2006).

Delta log-likelihood is calculated over held out data as follows:

$$\begin{aligned} \Delta_{\text{LL}} = \frac{1}{N} \sum_{n=1}^{N} \Big[\log p_{\text{model}} \left(\text{RT}_n\right) \\ - \log p_{\text{baseline}} (\text{RT}_n) \Big]. \end{aligned} \tag{34}$$

Language	Participants	Word Types	Observations	Train Size	Test Size
Dutch	45	1009	45314	40783	4531
English	46	980	57883	52095	5788
Finnish	49	1076	51962	46766	5196
German	45	1044	58707	52836	5871
Greek	45	975	41229	37106	4123
Hebrew	47	1319	50126	45113	5013
Italian	54	1071	62485	56237	6248
Korean	32	1249	21702	19532	2170
Spanish	48	1045	58554	52699	5855
Turkish	29	1056	23440	21096	2344
Russian	46	1194	48066	43259	4807

Table 7: Per-language counts for the Priority MMC (Approx.) estimator: observations entering the gaze-duration model are per participant. Train/test split refer to the size of the folds used to fit and test each of the GAMMs.

F.3 Estimator Configurations

Here we provide implementation details on the six sets of unigram surprisals compared in §8. The `wordfreq` values (Speer, 2022) are the corpus-based estimates used by Opedal et al. (2024). The NMC estimates (Eq. (11)) are computed from a large pool of $N = 4 \times 10^7$ samples, in order to circumvent the issue of poor coverage documented in §E. The MMC estimates (Eq. (19)) and their canonical-only approximations (Eq. (20); MMC and MMC Approx.) are fit over $N = 500$ samples. The priority-MMC estimates and their canonical-only approximations (Priority MMC and Priority MMC Approx.) use language-specific strata drawn from a pool of $N = 4 \times 10^7$ samples to balance language coverage; see §I.

G Variation across Languages

Where we report Δ_{LL} averaged across languages (e.g., Fig. 3), we specifically report micro-averaged Δ_{LL} over all of the word-level observations, pooled across the 11 languages. This means that languages with more data (see §F) contribute more to the means. Throughout this section, we present the macro-averaged Δ_{LL} and their standard deviations for the models of gaze duration. Values are presented separately for each estimator, corresponding to each of the facets in Fig. 3.

Unigram Surprisal Metric	Model	Macro Averages	SD Across Languages
NMC Estimate	Uni. Surp.	0.0265	0.0114
	PMI	0.0071	0.0062
	PMI & Uni. Surp.	0.0286	0.0121
	Con. Surp.	0.0118	0.0049
	Con. Surp. & Uni. Surp.	0.0283	0.0119
Corpus Estimate	Uni. Surp.	0.0282	0.0126
	PMI	0.0076	0.0057
	PMI & Uni. Surp.	0.0304	0.0132
	Con. Surp.	0.0118	0.0049
	Con. Surp. & Uni. Surp.	0.0299	0.0128

Table 8: Word-level ΔLL for GAMMs of gaze duration, macro-averaged across languages. Values correspond to the NMC estimator versus the Corpus baseline (fit on NMC's support).

Unigram Surprisal Metric	Model	Macro Averages	SD Across Languages
MMC Estimate	Uni. Surp.	0.0383	0.0195
	PMI	0.0122	0.0085
	PMI & Uni. Surp.	0.0415	0.0205
	Con. Surp.	0.0211	0.0100
	Con. Surp. & Uni. Surp.	0.0413	0.0204
Corpus Estimate	Uni. Surp.	0.0396	0.0211
	PMI	0.0103	0.0070
	PMI & Uni. Surp.	0.0430	0.0216
	Con. Surp.	0.0211	0.0100
	Con. Surp. & Uni. Surp.	0.0426	0.0214

Table 9: Word-level ΔLL for GAMMs of gaze duration, macro-averaged across languages. Values correspond to the MMC estimator versus the Corpus baseline (fit on MMC's support).

Unigram Surprisal Metric	Model	Macro Averages	SD Across Languages
MMC (Approx.) Estimate	Uni. Surp.	0.0391	0.0178
	PMI	0.0130	0.0085
	PMI & Uni. Surp.	0.0424	0.0191
	Con. Surp.	0.0214	0.0105
	Con. Surp. & Uni. Surp.	0.0421	0.0191
Corpus Estimate	Uni. Surp.	0.0396	0.0211
	PMI	0.0101	0.0069
	PMI & Uni. Surp.	0.0433	0.0217
	Con. Surp.	0.0214	0.0105
	Con. Surp. & Uni. Surp.	0.0429	0.0215

Table 10: Word-level ΔLL for GAMMs of gaze duration, macro-averaged across languages. Values correspond to the MMC (Approx.) estimator versus the Corpus baseline (fit on MMC (Approx.)'s support).

Unigram Surprisal Metric	Model	Macro Averages	SD Across Languages
Priority MMC Estimate	Uni. Surp.	0.0392	0.0175
	PMI	0.0111	0.0078
	PMI & Uni. Surp.	0.0421	0.0184
	Con. Surp.	0.0211	0.0104
	Con. Surp. & Uni. Surp.	0.0416	0.0184
Corpus Estimate	Uni. Surp.	0.0387	0.0198
	PMI	0.0104	0.0069
	PMI & Uni. Surp.	0.0426	0.0204
	Con. Surp.	0.0211	0.0104
	Con. Surp. & Uni. Surp.	0.0422	0.0202

Table 11: Word-level ΔLL for GAMMs of gaze duration, macro-averaged across languages. Values correspond to the Priority MMC estimator versus the Corpus baseline (fit on Priority MMC's support).

Unigram Surprisal Metric	Model	Macro Averages	SD Across Languages
Priority MMC (Approx.) Estimate	Uni. Surp.	0.0412	0.0201
	PMI	0.0110	0.0087
	PMI & Uni. Surp.	0.0436	0.0208
	Con. Surp.	0.0214	0.0105
	Con. Surp. & Uni. Surp.	0.0433	0.0207
Corpus Estimate	Uni. Surp.	0.0397	0.0211
	PMI	0.0101	0.0069
	PMI & Uni. Surp.	0.0433	0.0217
	Con. Surp.	0.0214	0.0105
	Con. Surp. & Uni. Surp.	0.0429	0.0216

Table 12: Word-level ΔLL for GAMMs of gaze duration, macro-averaged across languages: Values correspond to the Priority MMC (Approx.) estimator versus the Corpus baseline (fit on Priority MMC (Approx.)'s support).

H Estimator Pseudocode

Algorithm 1 Naive Monte Carlo Unigram Probability Estimator

Require: Word w, samples $X_1, \ldots, X_N$ where each $X_n = (x_1, \ldots, x_{L_n})$ is a token string

total Count $\leftarrow 0$
total Words $\leftarrow 0$
for each n in range N **do**
    total Count $\leftarrow$ total Count $+ \#(X_n, w)$
    total Words $\leftarrow$ total Words $+ |\pi(X_n)|$
end for
return $\dfrac{\text{total Count}}{\text{total Words}}$

Algorithm 2 Marginalized Monte Carlo Unigram Probability Estimator (Canonical-Only Variant)

Require: Word w, samples $X_1, \ldots, X_N$ where each $X_n = (x_1, \ldots, x_{L_n})$ is a token string

$[t_1, \ldots, t_T] \leftarrow \text{tokenize}(w)$
total Prob $\leftarrow 0$
total Words $\leftarrow 0$
for each n in range N **do**
  $B_n \leftarrow \{0\} \cup \{b\colon x_{b+1} \text{ is unit-initial}\}$
  total Words $\leftarrow$ total Words $+ |B_n|$
  for each boundary position $b \in B_n$ **do**
    prefix $\leftarrow (x_1, \ldots, x_b)$
    prob $\leftarrow \vec{p_\Sigma}(t_1 \mid \text{prefix})$
    for $i = 2, \ldots, T$ **do**
      prob $\leftarrow$ prob $\cdot\, \vec{p_\Sigma}(t_i \mid \text{prefix} \circ [t_1, \ldots, t_{i-1}])$
    end for
    total Prob $\leftarrow$ total Prob $+$ prob
  end for
end for
return $\dfrac{\text{total Prob}}{\text{total Words}}$

I Priority-Sampled Evaluation at Large Sample Sizes

Alg. 2 evaluates $\vec{p_\Sigma}$ at every boundary of every sample. A unit whose probability mass lies in contexts that are rare under p_Σ requires a large sample size N before any such context is drawn. Sampling and the sweep that returns every unit's first-token probability are paid once, no matter how many units are estimated. However, a unit of T tokens requires $T-1$ forward passes at each boundary, computed separately for each unit and linear in the pool size (Tab. 2). We therefore keep the sweep exhaustive and evaluate the continuation only at a bounded set of boundaries, chosen via priority sampling (Duffield et al., 2007). We present two variants: a canonical-only variant (Alg. 3; Priority MMC Approx. in §8), and a full estimator (Priority MMC), which restores the exact-count EOS correction of Eq. (16).

Let $\mathcal{F}$ collect every boundary position of every sample. For a word w with tokenization $[t_1, \ldots, t_T]$ we write

$$s_b = \vec{p_\Sigma}\left(t_1 \mid \text{prefix}_b\right), \qquad c_b = \prod_{i=2}^{T} \vec{p_\Sigma}\left(t_i \mid \text{prefix}_b \circ [t_1, \ldots, t_{i-1}]\right), \tag{35}$$

with $c_b = 1$ when $T = 1$. Summing these terms and normalizing, $\sum_{b \in \mathcal{F}} s_b c_b / \ell$, gives the canonical-only variant, writing ℓ for the realized word count of the pool, which approximates Eq. (19): (i) the summands are evaluated only at positions whose sampled successor is unit-initial, (ii) c_b omits the right-boundary mass of §C, and (iii) residual occurrences are dropped (Eq. (20)). The full estimator below removes these three approximations.

Proposition 3. *Fix a stratum g and condition on the pool. Then*

$$\mathbb{E}\left[\sum_{b \in \mathcal{S}_g} \max(s_b, \tau_g)\, c_b\right] = \sum_{b \in \mathcal{F}_g} s_b c_b \tag{36}$$

Proof. If $|\mathcal{F}_g| \le k_g$ then $\mathcal{S}_g = \mathcal{F}_g$ and $\tau_g = 0$, so the two sides agree term by term. Otherwise fix b and let τ_{-b} be the k_g-th largest priority among $\mathcal{F}_g \setminus \{b\}$, a function of $\{u_{b'}\}_{b' \neq b}$ and so independent of u_b. Boundary b is retained exactly when $q_b > \tau_{-b}$, and then the (k_g+1)-th largest priority overall is τ_{-b}, so

$$\mathbb{E}[\max(s_b, \tau_g)\, c_b \mathbb{1}\{b \in \mathcal{S}_g\} \mid \tau_{-b}] = \max(s_b, \tau_{-b}) \min\left(1, \tfrac{s_b}{\tau_{-b}}\right) c_b = s_b c_b, \tag{37}$$

Algorithm 3 Priority-Sampled Marginalized Monte Carlo Estimator (canonical-only variant)

Require: Word w, frame $\mathcal{F}$ partitioned into strata $\{\mathcal{F}_g\}$, budgets $\{k_g\}$, pool word count ℓ, first-token probabilities s_b from the sweep

$[t_1, \dots, t_T] \leftarrow \text{tokenize}(w)$
if $T = 1$ **then**
  return $\sum_{b \in \mathcal{F}} s_b \,/\, \ell$
end if
total $\leftarrow 0$
for each stratum g **do**
  $q_b \leftarrow s_b / u_b$ **for** $b \in \mathcal{F}_g$, **with** $u_b \sim \text{Uniform}(0, 1]$
  $\mathcal{S}_g \leftarrow$ the k_g boundaries of $\mathcal{F}_g$ with largest q_b
  $\tau_g \leftarrow$ the $(k_g + 1)$-th largest q_b, or 0 if $|\mathcal{F}_g| \le k_g$
  for each $b \in \mathcal{S}_g$ **do**
    $c_b \leftarrow 1$
    for $i = 2, \dots, T$ **do**
      $c_b \leftarrow c_b \cdot \vec{p_\Sigma}(t_i \mid \text{prefix}_b \circ [t_1, \dots, t_{i-1}])$
    end for
    total $\leftarrow$ total $+ \max(s_b, \tau_g) \cdot c_b$
  end for
end for
return total $/\, \ell$

by $\mathbb{P}[u_b < s_b/\tau_{-b} \mid \tau_{-b}] = \min(1, s_b/\tau_{-b})$ and $\max(w, \tau)\min(1, w/\tau) = w$. The right-hand side does not involve τ_{-b}, so averaging over it leaves $\mathbb{E}[\max(s_b, \tau_g)\, c_b \mathbb{1}\{b \in \mathcal{S}_g\}] = s_b c_b$. Since

$$\sum_{b \in \mathcal{S}_g} \max(s_b, \tau_g)\, c_b = \sum_{b \in \mathcal{F}_g} \max(s_b, \tau_g)\, c_b \mathbb{1}\{b \in \mathcal{S}_g\}\,, \tag{38}$$

summing this expectation over $b \in \mathcal{F}_g$ gives the claim by linearity of expectation. □

Stratification. We form strata from each boundary's own prefix and refine the Latin script class with language detection. Proposition 3 holds within each stratum, so the sum is unbiased for any allocation $(k_g)_g$ and the allocation may be chosen to reduce variance. Two conditions are needed: every stratum carrying mass must receive $k_g \ge 1$, since an unfunded stratum contributes nothing while its mass stays in the estimand, and the allocation must not depend on the draws, which holds because it comes from the column sums. Labels therefore enter the allocation and never a weight, so a mislabeled boundary costs precision but cannot bias the estimate.

Full estimator. The full estimator removes the three approximations in Alg. 3:

1. An interior position enters $\mathcal{F}$ only when its sampled successor is unit-initial, an event of probability W_b, the unit-initial mass of $\vec{p_\Sigma}(\cdot \mid \text{prefix}_b)$, so $s_b = \vec{p_\Sigma}(t_1 \mid \text{prefix}_b)\,/W_b$.

2. The continuation c_b adds the right-boundary factor of §C, i.e., the mass on unit-initial tokens and EOS after t_T.

3. Residual occurrences are recovered through C, which is the exact count of the unit over the pool, taken on the decoded strings so that every tokenization contributes.

Writing $\mathcal{T} \stackrel{\text{def}}{=} \{b \in \mathcal{F}\colon$ the sampled successor at b is $t_1\}$, the full estimator is $\widehat{N}(w) \stackrel{\text{def}}{=} \widehat{S}(w) + C - \widehat{H}(w)$, where $\widehat{S}$ priority-samples $\sum_b s_b c_b$ over the interior boundaries and adds the exact empty-prefix term, common to all N samples, and $\widehat{H}$ estimates $\sum_{b \in \mathcal{T}} c_b$ from at most k_{take} positions of $\mathcal{T}$ drawn uniformly per first token, each weighted by the ratio of available to drawn positions. Since the two share the

summand c_b, the tower property gives $\mathbb{E}\left[\widehat{S}\right] = \mathbb{E}\left[\widehat{H}\right]$, so $\widehat{N}$ is unbiased for N times the expected count of Eq. (7) under the length-truncated sampling process of §7, and the reported estimate is $\widehat{N}/\ell$. A negative realized numerator, which occurs for units far below the pool's resolution, leaves the estimate undefined and the unit excluded. Unbiasedness refers to $\widehat{N}$ before this exclusion.

Settings. Tab. 13 shows the sampling parameters of both variants. We allocated the per-unit budget k in three steps: we divide a fixed share among the strata of the languages in which the unit occurs, half of it going to a generic Latin stratum for languages written in Latin script. Every stratum that carries mass receives a floor of 64 boundaries, the rest is divided in proportion to each stratum's share of the unit's first-token mass, capped at the stratum's total from the first two steps. Any residue is divided by mass alone, such that the full budget is spent whenever the stored cells can serve it.

	Approx.	Full
Samples N	4×10^7	
Maximum length	50	
Strata	21	
Per-unit budget k	8192	
Stratum floor	64	
Home share	4096	2048
Draw seed	100000	900000
Frame size $\lvert\mathcal{F}\rvert$	8.9×10^8	9.2×10^8
Take draws k_{take}	n/a	2048
Single-token units	exact	sampled

Table 13: Settings of the priority-sampled variants (Priority MMC Approx. and Priority MMC). Each variant draws its own pool of samples. The full-variant's pool additionally stores the model's own sampled tokenizations for $\widehat{H}$. Approximate-variant single-token units need no sampling, since their numerator is the exact column sum plus the exact empty-prefix term. The full estimator's closing mass depends on the unit, so its single-token units are sampled like any other.